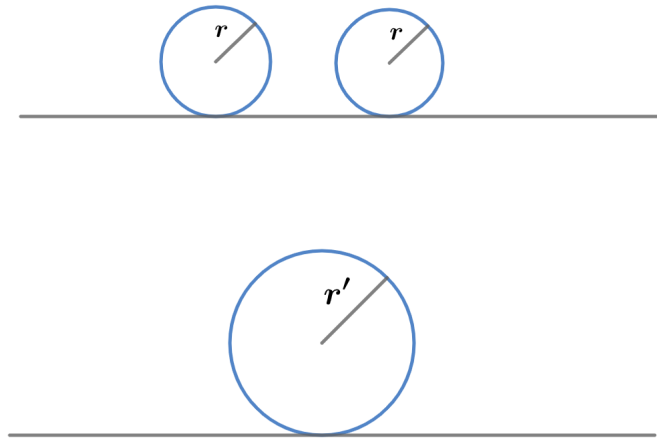


2020A F=ma Exam: Problem 16

Kevin S. Huang



Since volume $V \propto r^3$, we have $2r^3 = r'^3$ by conservation of mass which means $r' \propto r$. Since $U \propto r^2$ and $r' \propto r$, we also have $\Delta U \propto r^2$. By conservation of energy,

$$\Delta U = mgh$$

Since $m \propto r^3$, the above equation becomes

$$r^2 \propto r^3 h$$

Thus,

$$h \propto \frac{1}{r}$$

so the answer is D.